Philosophy of Science
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Lecture 5
Falsificationism

Where in the course are we?

I. THE BASICS

1. Ways of knowing
2. Knowledge, truth, and facts

II. SOME SCIENTIFIC DEVICES

3. Scientific laws and induction
4. Explanations and causes

III. MODELS OF SCIENTIFIC PRACTICE

5. Falsificationism ← *Today*
6. Paradigms and revolutions

Motivation of this lecture

- Why do you need to know about falsificationism? Because…
  - It is an insightful suggestion about how science works
  - Many scientists and scholars think that it is the key to scientific knowledge
- You will often hear, “Interesting theory—but is it falsifiable?”
  - You need to know how to respond to this

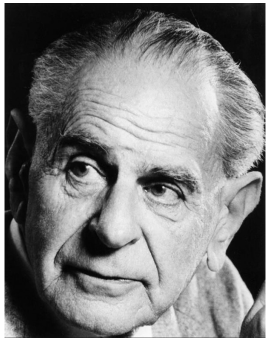
Plan of this lecture

- Introducing Karl R. Popper
- Falsificationism: basic principles
  - Asymmetry of confirmation and refutation
  - Rejection of induction
  - Distinction discovery / justification
  - Demarcation science / non-science
- Critical appraisal of falsificationism

Reading for this lecture

- Philip Parvin, *Karl Popper*
  - New York: Continuum, 2010
  - Read chapter 2, "Popper's Ideas", first two sections, pp. 33–66
- Available on Brightspace

Karl R. Popper, 1902–1994



- Born in Vienna
- PhD in philosophy, Vienna, 1928
- Taught in New Zealand, 1937–1945
- Professor at London School of Economics from 1945 onwards
- Founded school of "critical rationalism"

Popper's principal works

- *The Open Society and Its Enemies*, 1945
  - Liberalist polemic against Plato, Marx, and others who believed in rigid social orders and historical laws
- *The Logic of Scientific Discovery*, 1959
  - Treatise in which Popper unveiled falsificationism
- *Conjectures and Refutations*, 1963
  - Collection of essays on science as trial and error
- *Objective Knowledge*, 1972
  - Drew link between science and Darwinian evolution
- *Unended Quest*, 1982
  - Insightful intellectual autobiography

Glossary

- To falsify: to refute, i.e. to establish empirically that a hypothesis is untrue
- Falsifiability: the susceptibility of a hypothesis to being falsified
- Falsification: the act of falsifying a hypothesis
- Falsificationism: Popper's thesis—scientific method consists in falsification

Popper's concerns

- Popper posed the following questions about scientific method:
  - What is a sound way to test scientific hypotheses?
  - How can models of scientific practice do justice to the creativity of scientists?
  - How does science differ from non-science?
- Overriding priority
  - Eliminating uncertainty from science

How to test hypotheses 1

- In science, we need to test hypotheses
  - Usually, we cannot check a hypothesis in its entirety—it is too general or too abstract
  - But we can check its observational implications or predictions
- Our strategy:
  - Hypothesis *H* entails observational claim *O*
  - We check whether *O* is true or false
  - We then draw an inference about the adequacy of *H*

How to test hypotheses 2

- Example:

  Newtonian model of the solar system (*H*)

  entails

  There will be an eclipse on 1 May 2030 (*O*)
- We cannot test the entire Newtonian model directly
  - But we can test observational claim *O*
  - … by ascertaining whether an eclipse occurs on 1 May 2030

How to test hypotheses 3

- The test of an observational claim *O* has two possible outcomes:
  - We observe *O* to be true
  - We observe *O* to be false
- Popper's crucial insight
  - There is an important asymmetry between these two outcomes

Outcome 1: *O* is true

- We observe *O* to be true
  - An eclipse occurs on 1 May 2030
- It is still possible that *H* is false
  - Some other mechanism may have led to *O*
  - An eclipse may occur on 1 May 2030 even if the Newtonian model is false
- Thus, observing that *O* is true gives us no certainty that *H* is true

Outcome 2: *O* is false

- We observe *O* to be false
  - No eclipse occurs on 1 May 2030
- Then, it must be that *H* is false
  - If the mechanism posited by *H* were real, then it would certainly lead to *O*
  - If no eclipse occurs on 1 May 2030, then the Newtonian model must be false
- Thus, the single observation that *O* is false gives us certainty that *H* is false

Asymmetry of hypothesis test

- Any attempt to…
  - confirm a hypothesis by showing that it entails true observational claims fails to deliver certainty
  - refute a hypothesis by showing that it entails false observational claims succeeds in delivering certainty
- Thus, we should pursue only refutation
  - … and abandon confirmation of hypotheses

Underlying logical schemas

- Confirmation depends on an invalid deductive argument:

  H entails *O*
  O
  ——————
  Hence, *H* ← Logically invalid

- Refutation uses a valid deductive argument:

  H entails *O*
  Not *O*
  ———————
  Hence, not *H* ← Logically valid

Popper's rejection of induction

- By rejecting confirmation, Popper rejected induction
- Induction is a pattern of reasoning…
  - from observations of individual instances to generalisations—see lecture 3
- Any confirmation is an inductive process
  - Popper thought that David Hume's problem of induction was unsolvable
  - He campaigned to stamp out induction in scientific practice

Science according to Popper 1

- Scientists can attempt only to falsify hypotheses
- Two possible results of hypothesis test:
  - Hypothesis is falsified, and scientists must abandon it at once
  - Hypothesis is not falsified: it survives to the next test—but receives no confirmation
- Method of trial and error
  - Attractively undogmatic picture of science

Science according to Popper 2

- What does scientific knowledge consist in?
  - A set of not yet refuted hypotheses
  - Not a set of true or highly confirmed findings
- What does scientific progress consist in?
  - Elimination of errors
  - Not accumulation of truths
- Characterisation of scientific practice in "negative" terms

Origin of hypotheses

- Popper still has one question to answer
  - Where do hypotheses come from?
- Scientists invent hypotheses freely
  - They may use any source of inspiration: dreams, observations, random associations…
- No source of inspiration gives us any reason to believe a hypothesis, however
  - Not even observations—that would be induction!

Discovery and justification

- Two phases of scientific practice
- Context of discovery
  - Phase in which scientists freely conjecture hypotheses
  - No rules or standards: unbridled creativity
- Context of justification
  - Phase in which scientists test hypotheses
  - Phase of logic and rigour
  - Assures objectivity of science

Demarcation of science 1

- Popper also applied his view to the problem of recognising science
- If falsification is the scientific method…
  - … then it is also the hallmark of science
- Demarcation science / non-science
  - Practitioners of a discipline must be able to say what would falsify their hypotheses
  - … or what outcome a hypothesis rules out
  - Otherwise the discipline is a pseudo-science

Demarcation of science 2

- Example: astrology
  - "Next month you will meet a tall, mysterious stranger and your life will change"
- Astrological claims are unfalsifiable
  - That is why astrology is a pseudo-science
  - Scientific status follows from falsifiability, not from degree of confirmation of claims

Influence of Popper

- Falsificationism still has wide influence outside philosophy of science
- Many researchers think the crucial question is…
  - Interesting theory—but is it falsifiable?
- Some use falsifiability criterion as test of scientific status of disciplines
  - Is psychoanalysis scientific? Cultural studies? Economics?

Critical evaluation

- Four criticisms of falsificationism:
  - Refutation is less straightforward than Popper assumed
  - Lack of accord with scientific practice
  - Popper neglects possible ways of making confirmation more rigorous
  - Some valuable scientific hypotheses seem not to be falsifiable

Critique of falsificationism 1

- Refutation is less straightforward than Popper assumed
- Many hypotheses yield no predictions on their own
  - ... but only if they are combined with auxiliary assumptions
  - It becomes difficult to pin the blame for a failed prediction on any single hypothesis

Critique of falsificationism 2

- Lack of accord with scientific practice
  - Even Popper's heroes, such as Albert Einstein, were not rigorous falsificationists
  - Scientists make *ad hoc* modifications of hypotheses to avert refutation
  - This is often fruitful for scientific progress
- Falsification may be less relevant to scientific practice than Popper thought
  - Methodological value of dogmatism

Critique of falsificationism 3

- Popper neglects possible ways of making confirmation more rigorous
  - Pragmatic warrant of induction—see lecture 3
  - Bayesian confirmation theory
- These allow a more "positive" characterisation of scientific practice
  - Even Popper, at the end of his career, acknowledged "corroboration", a weak form of confirmation

Critique of falsificationism 4

- The sole falsifiable hypotheses are…
  - *A* is a necessary or sufficient condition for *B*
- Probabilistic hypotheses are not strictly falsifiable by one or a few instances
  - *A* is statistically correlated with *B*
  - Example: income and health
- Therefore, criterion of falsifiability may rule out examples of good science
  - Especially in social science

Verdict on falsificationism

- For:
  - Simple, logical model of scientific method
  - Attractive image of scientists as creative, risk taking, undogmatic
- Against:
  - Narrow view of scientific practice
  - Underestimates complexity of observation and theory test
  - Neglects mechanisms of confirmation and empirical support

Recapitulation

- Popper and falsificationism
  - Asymmetry of confirmation and refutation
  - Rejection of induction
  - Demarcation science / non-science
- Appraisal and critique
  - Most philosophers no longer accept falsificationism as the best model of science
  - But many scientists and scholars still take falsification as the way science works